1. virtual environments
part-1 : after installing python in your system install numpy package but install version 2.3.5 specifically in your system (systemwide first, not in venv). Now do “pip list” to list all python packages in your system. Locate numpy and run the following python script

```
import numpy as np
print(np.__version__)
```

part-2 : create a empty folder and inside, create a virtual environment named “v1”. Enter the environment and install numpy package again (this time just do pip install numpy, to install latest default numpy version) and run the same script from inside the environment. And note the output

note : inside the env we can set it in such a way that it will use a specific version (if installed). Not done here, just for your information

2. using numpy

create a python list of size 15 and fill it with random integer numbers between [10,20] (use random module). Now create a numpy array X from it. Create following numpy arrays from the first one
(i) create a numpy array Y which contains list of y[i] = 2*x[i] + 3. y coordinate of point on the line y=2x+3 where x coordinate comes from the X list
(ii) using numpy methods. Print the size of X and max, min, avg of elements in X
(iii) print the type of the initial list and the type of the numpy array

3. using numpy

create 2 multidimensional lists of shape 3x3 X,Y. fill them with random integer numbers between [10,20] (use random module).
(i) convert them both into numpy ndarrays nX, nY respectively. Now do Z1 = nX*nY and Z2 = nX@nY. Explain the difference
(ii) Do nX = nX/10. And use log, exp, sqrt functions in numpy to print the log of every element, e^ every element, square root of every element of nX in a matrix form directly

4. using pandas

use the hdp1.csv file I gave before. Using read_csv functino in pandas , load the data from the file into a dataframe object. Also use head and tail functions and print the output

From the dataframe object print details of the dataset like shape, no of rows, columns. Index list, columns names list along with datatypes of all columns

5. pandas – row and column manipulation

load the hdp1.csv into a dataframe object *df* and do the following
(i) delete the last column in the dataset
(ii) add a new column named A which is just 2.B + 3 where B is any numerical column you have in the dataset